

Differential CC $1\mu 1\pi^\pm$ cross sections on argon in the NuMI beam

Inclusive measurement and the proton-tagged hadronic-mass / TKI subsample

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$\nu_\mu + \bar{\nu}_\mu$ charged-current single charged pion • FHC, RHC, combined

Part I — Inclusive $CC1\mu1\pi^\pm Xp$

1. Motivation & signal
2. Beam, exposure, selection
3. Cut-flow & performance
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Part II — Proton-tagged (W / TKI)

1. Motivation: hadronic mass & TKI
2. Selection & proton-tag performance
3. Observables & closure
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Signal definition

$\nu_\mu/\bar{\nu}_\mu$ charged-current, exactly one muon and one charged pion, any number of protons, no π^0 /kaons/heavier mesons; muon $p_\mu > 0.15$, pion $p_\pi > 0.175$ GeV/c, in the fiducial volume.

- ▶ Resonant $\Delta(1232)$ production dominates; probes pion production, nuclear effects and final-state interactions on argon.
- ▶ NuMI off-axis beam: ν_μ -dominant FHC, $\bar{\nu}_\mu$ -enriched RHC.
- ▶ Five differential observables: p_μ , p_π , $\cos\theta_\mu$, $\cos\theta_\pi$, $\theta_{\mu\pi}$.

Exposure (fake data, blind)

FHC (Runs 1,2,4,5)	8.86×10^{20} POT
RHC (Runs 1,2,3,4)	1.11×10^{21} POT
Combined	1.99×10^{21} POT

Event selection: cut sequence

A $CC1\mu1\pi^\pm Xp$ topological selection: one muon, exactly one contained charged pion, any number of protons. Applied in order (cosmic rejection first, then PID, then kinematic cleanup):

1. **NeutrinoSlice** — ≥ 1 track, software trigger.
2. **InFiducialVol** — SCE-corrected ν vertex in FV.
3. **Topological** — Pandora score > 0.67 ; primary cosmic rejection (EXT $\times 33$ down).
4. **MuonCandidate** — ≥ 1 primary track passes μ PID; longest = μ .
5. **ContainedPion** — exactly one other track passes π PID, endpoint contained; highest LLR = π .
- 6–7. **μ/π In3Planes** — hit on each of U, V, Y.
8. **ShowerCut** — veto primary EM showers (π^0 rejection; one soft shower tolerated).
9. **OpeningAngle** — $\theta_{\mu\pi} < 2.6$ rad (149°); rejects large-angle cosmic/dirt.
10. **Nonprotons** (final) — < 4 non-proton-like tracks (LLR > 0.1) and < 5 total primary tracks.

Per-run POT scaling; EXT (beam-off) supplies data-driven cosmics; dirt from the GENIE NuMI overlay. **Final: efficiency** 15.6%, **purity** 55% (measured phase space).

Particle identification (muon & pion candidates)

Track-level PID combines Pandora track score, LLR calorimetric PID, Bragg-peak likelihoods and dedicated MIP/pion BDTs:

Muon candidate (Cut 4)

Variable	Threshold
Track score	> 0.80
Vertex distance	≤ 4.0 cm
Track length	> 10.0 cm
LLR PID	> 0.20
MIP BDT	≥ -0.10

Pion candidate (Cut 5)

Variable	Threshold
LLR PID	> 0.10
Vertex distance	< 4.0 cm
Track length	> 20.0 cm
MIP BDT	> -0.10
Pion BDT (vs. p)	> -0.10
Bragg-pion score	≥ 0.08
Endpoint contained	yes

Pion momentum from the proton-hypothesis range KE (tracks low-energy stopping pions better than the μ hypothesis).

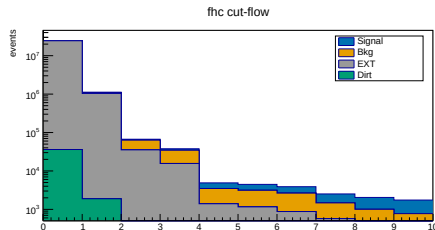
Inclusive cut-flow — FHC (full exposure)

Cut	Signal	Bkg	EXT	Eff%	Pur%
None	5 580	279 180	24.4M	100.0	0.02
InFiducialVol	4 632	62 146	1.04M	83.0	0.42
Topological	3 501	28 040	35 184	62.7	5.2
MuonCandidate	3 143	18 980	15 516	56.3	8.3
ContainedPion	1 395	2 094	1 376	25.0	28.5
MuonIn3Planes	1 336	1 975	1 145	23.9	29.9
PionIn3Planes	1 239	1 770	872	22.2	31.9
ShowerCut	1 048	898	569	18.8	41.6
OpeningAngle	1 040	813	190	18.6	50.9
Nonprotons	974	643	131	17.5	55.7

MC prediction, per-run POT scaled; data blind. Dirt $\lesssim 1$ event at final cut.

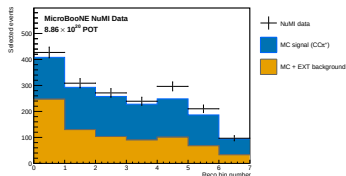
Final cut (Nonprotons), all beams:

Beam	Signal	Bkg	EXT	Eff%	Pur%
FHC	974	643	131	17.5	55.7
RHC	1 081	746	136	17.6	55.0
Comb	2 055	1 389	266	17.5	55.4

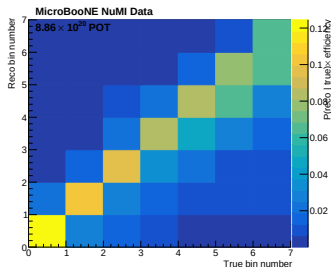


FHC stack (signal blue, bkg orange, EXT grey, dirt green; log scale). Two big drops: Topological (cosmics) and ContainedPion (single- π topology).

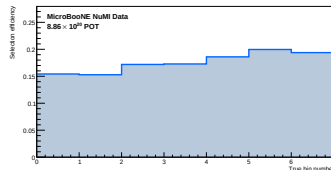
Reconstruction, response & efficiency



Reco p_μ (data/MC stack)



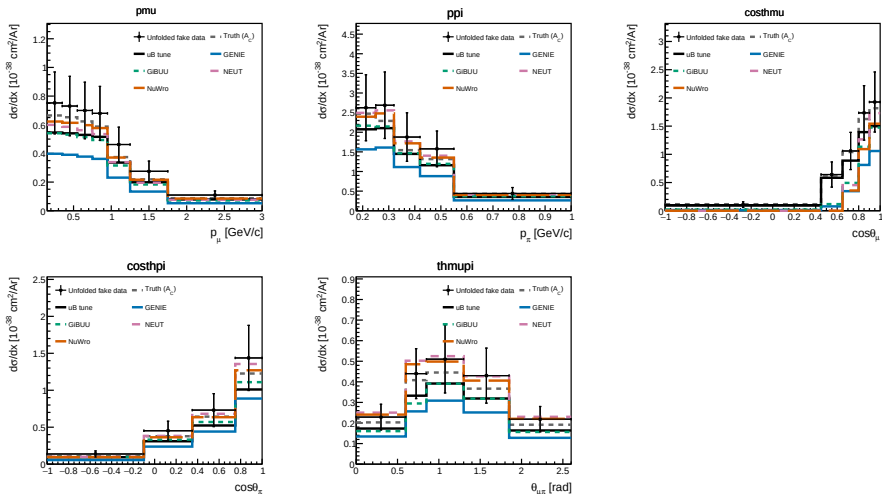
Response matrix



Efficiency vs p_μ

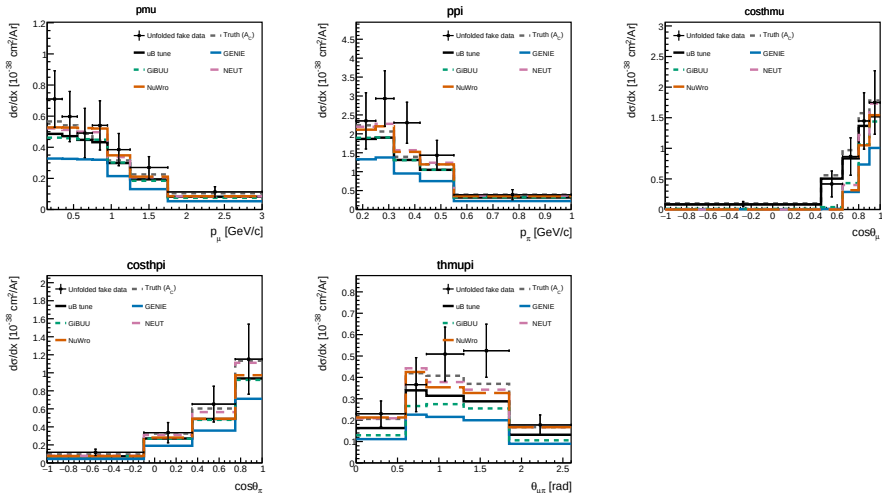
Wiener-SVD unfolding (framework WienerSVDUnfolder, 2nd-derivative regularisation) with the full systematic covariance from the universe throws.

Differential cross sections — FHC



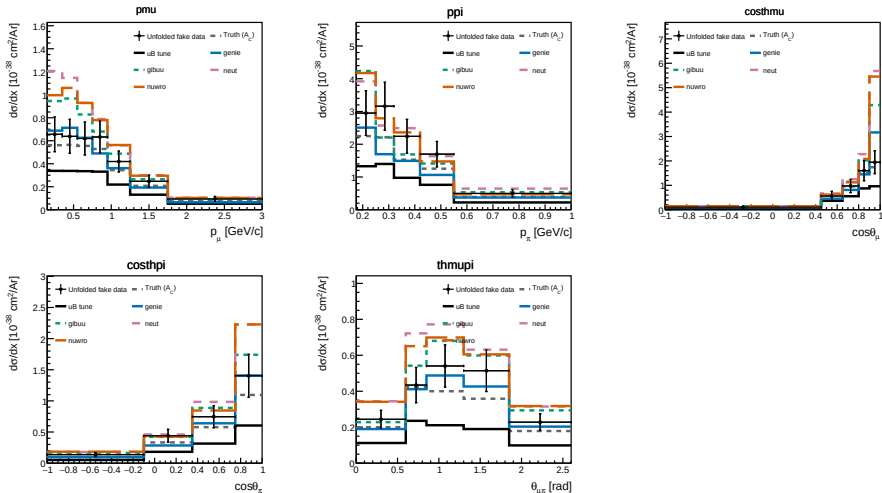
Unfolded fake data vs A_C -smeared truth, MicroBooNE tune, and GENIE/GiBUU/NEUT/NuWro.

Differential cross sections — RHC



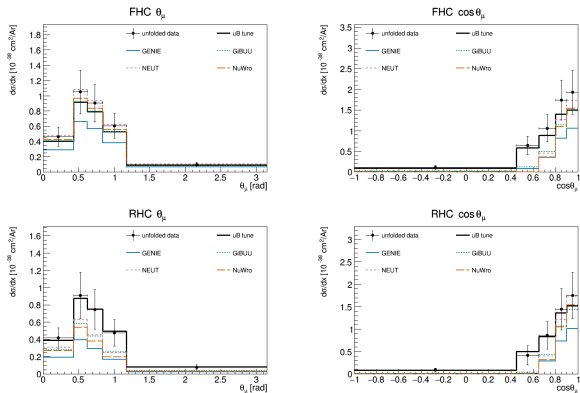
$\bar{\nu}_\mu$ -enriched beam; unfolded fake data vs A_C -smeared truth, MicroBooNE tune, and GENIE/GiBUU/NEUT/NuWro.

Differential cross sections — combined



FHC+RHC combined exposure (per-universe correlated flux treatment).

Muon angle: $\cos \theta_\mu$ vs θ_μ



All curves A_C -smeared (same space as the data). Same events, same binning (θ edges = arccos of the $\cos \theta$ edges) — only the variable differs.

Why θ_μ : the A_C smearing penalises the forward peak

Beam	Var	data	tune	GENIE	NuWro
FHC	$\cos \theta_\mu$	0.82	0.67	0.27	0.30
FHC	θ_μ	1.00	0.87	0.63	0.92
RHC	$\cos \theta_\mu$	0.66	0.63	0.23	0.26
RHC	θ_μ	0.83	0.82	0.33	0.42

A_C suppression (smeared/truth):

FHC $\cos \theta_\mu$ 0.33 $\rightarrow$ θ_μ 0.76 (p_μ 0.64)

RHC $\cos \theta_\mu$ 0.34 $\rightarrow$ θ_μ 0.49 (p_μ 0.72)

- ▶ $\cos \theta_\mu$ is sharply forward-peaked; the regularised measurement cannot resolve it, so A_C smears it away — penalising the **peaked generators** far more than the smoother tune.
- ▶ In θ_μ the $\sin \theta$ Jacobian flattens the peak $\Rightarrow$ suppression 0.33 $\rightarrow$ 0.76, generators land **on** the data.
- ▶ Not a model disagreement: truth-level generator totals agree; it is a **variable-choice artefact**.
- ▶ RHC improves less (poorer angular response, stronger regularisation). $\theta_\pi \approx \cos \theta_\pi$ (pion never sharply peaked).

Integrated & total flux-averaged cross section

Integrated σ_{int} (Wiener-SVD)

	FHC	RHC	Comb
p_μ	0.99	0.86	0.82
p_π	0.96	0.97	0.96
$\cos \theta_\mu$	0.82	0.66	0.70
$\cos \theta_\pi$	0.97	0.81	0.86
$\theta_{\mu\pi}$	0.88	0.88	0.92
θ_μ	1.00	0.83	—
θ_π	0.97	0.80	—

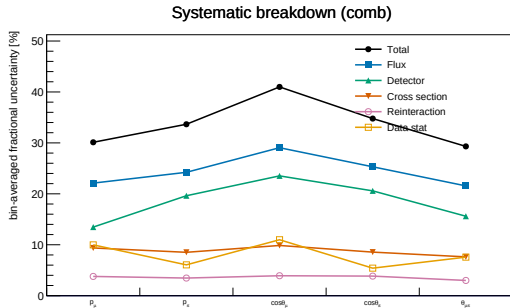
spread = Wiener-SVD A_C smearing ($10^{-38}\text{cm}^2/\text{Ar}$)

Total (cut-and-count, no unfolding)

Config	σ ($10^{-38}\text{cm}^2/\text{Ar}$)
FHC	1.06 ± 0.27
RHC	0.98 ± 0.35
Combined	1.02 ± 0.30

FHC generators: GENIE 0.79, GiBUU 1.08, NEUT 1.26, NuWro 1.20 — agree within $\sim 1\sigma$.

Systematics & sideband validation



Flux (PPFX) dominates; detector second; total $\sim 30\text{--}40\%$.

Background-control sidebands

Four signal-depleted regions ($CC0\pi$, multi- π , π^0 , cosmic). High-statistics data/ ν MC = 1.00 across FHC/RHC/comb $\Rightarrow$ per-run weighting & POT machinery validated; ready to constrain the background at unblinding.

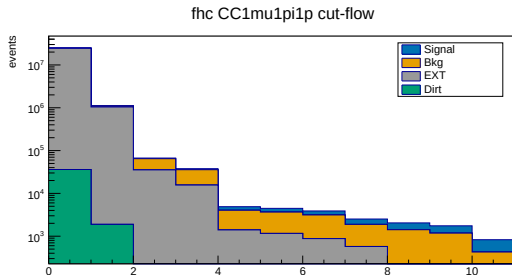
Part II — Proton-tagged subsample: hadronic mass & TKI

Idea

Requiring an identified proton reconstructs the full hadronic system, opening two new observable classes: the **hadronic invariant mass** (probes Δ/N^* directly) and the **transverse kinematic imbalance** (Fermi motion + FSI).

- ▶ CC1 μ 1p1p: inherits the inclusive selection + a leading proton (LLR PID, $p_p \geq 0.3$ GeV/ c , contained). Efficiency 11.5%, purity 48%.
- ▶ Six W/TKI observables: $W_{\pi p}$, W_{had} , δp_T , $\delta \alpha_T$, $\delta \phi_T$, p_n (6 bins each) + five inclusive kinematic observables re-measured on the proton-tagged phase space.

Selection performance & proton-tag quality



Proton tag (truth-matched, FHC)

- ▶ 87% tag purity ($\sim 13\%$ mis-tag)
- ▶ Rejects $\sim 64\%$ background for $\sim 30\%$ signal loss
- ▶ Purity $33\% \rightarrow 48\%$; $S/\sqrt{B} + 17\%$
- ▶ Proton p : -9% bias, 21% resolution

FHC proton-tagged cut-flow; last stage = proton identification

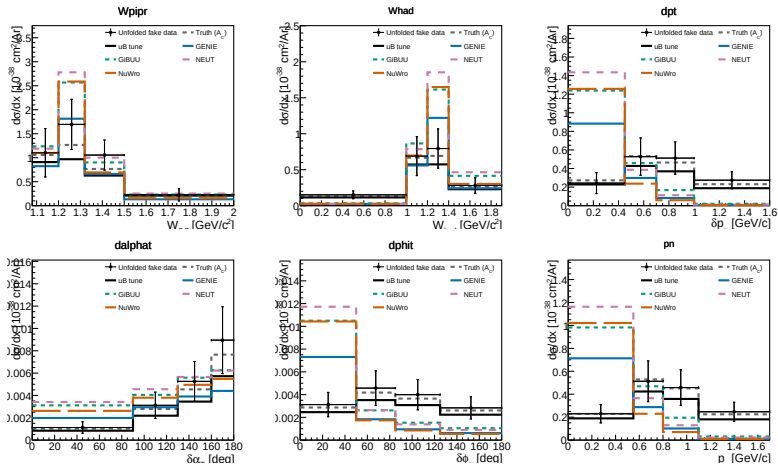
Proton-tagged cut-flow — FHC (full exposure)

Stages None–Nonprotons are the inherited inclusive selection (with the proton-tagged signal definition: inclusive signal + a true proton > 0.3 GeV/c); the final **IdentProton** stage adds the leading-proton identification that defines the subsample.

Cut	Signal	Bkg	EXT	Pred	Eff%	Pur%
None	3 387	281 372	24.4M	24.7M	100.0	0.01
InFiducialVol	2 834	63 944	1.04M	1.11M	83.7	0.26
Topological	2 238	29 303	35 184	66 929	66.1	3.3
MuonCandidate	2 031	20 092	15 516	37 776	60.0	5.4
ContainedPion	852	2 637	1 376	4 890	25.2	17.4
MuonIn3Planes	812	2 499	1 145	4 473	24.0	18.2
PionIn3Planes	754	2 255	872	3 888	22.3	19.4
ShowerCut	632	1 313	569	2 521	18.7	25.1
OpeningAngle	628	1 226	190	2 045	18.5	30.7
Nonprotons	568	1 049	130	1 749	16.8	32.5
IdentProton	401	373	53	828	11.8	48.4

IdentProton rejects $\sim 64\%$ of the ν background while keeping $\sim 71\%$ of signal: purity $32.5\% \rightarrow 48.4\%$. RHC/comb: 46.6%/47.5% purity at $\sim 12\%$ efficiency.

Proton-tagged differential cross sections (FHC)



Unfolded fake data vs truth, uB tune, and all four generators. Closure $\chi^2/\text{ndf} \ll 1$ for every observable.

Generator comparison & background composition

Integrated proton-tagged σ ($10^{-38}\text{cm}^2/\text{Ar}$)

GENIE 0.50 < NuWro 0.65 < GiBUU 0.74 < NEUT 0.78; data ~ 0.58 .

Spread $\sim 1.6\times$; largest in low- δp_T /high- p_n (FSI-sensitive).

NuWro/NEUT reprocessed from event vectors in the SL7 container.

Remaining background (FHC)

Component	%
ν_μ CC wrong topology	64
True CC1 μ 1 π , no $p>0.3$	18
NC	11
Out-of-FV	5
ν_e CC	1

82% true ν_μ CC — well modelled & constrainable.

Summary

- ▶ **Inclusive:** differential $\text{CC1}\mu 1\pi^\pm$ cross sections for FHC, RHC and combined, per-run POT scaling, Wiener-SVD unfolding; total flux-averaged σ agrees with GENIE/GiBUU/NEUT/NuWro within $\sim 1\sigma$.
- ▶ **Proton-tagged:** new hadronic-mass ($W_{\pi p}$, W_{had}) and TKI (δp_T , $\delta\alpha_T$, $\delta\phi_T$, p_n) observables, plus the five inclusive kinematic observables re-measured. **FHC closure validated** at six-bin granularity; **RHC not yet validated** (unfolds negative — background > data in this low-purity subsample), combined deferred behind it.
- ▶ **Blind analysis:** fake data throughout; sidebands ($\text{CC0}\pi$, multi- π , π^0 , cosmic) validated in all three beams, data/ ν MC $\simeq 1.00$, ready to constrain the background before unblinding.
- ▶ **Ongoing:** $\cos\theta \rightarrow \theta$ switch (the A_C smearing suppresses the forward-peaked $\cos\theta_\mu \times 0.33$ vs $\times 0.76$ in θ_μ); RHC proton-tagged closure; detector systematic; tighter-proton-PID study.

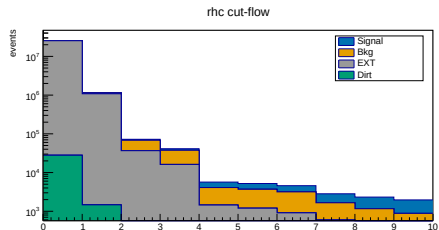
Thank you — questions?

Backup

Backup: inclusive cut-flow — RHC (full exposure)

Cut	Signal	Bkg	EXT	Eff%	Pur%
None	6 131	304 347	25.3M	100.0	0.02
InFiducialVol	5 124	69 102	1.08M	83.6	0.44
Topological	3 960	31 629	36 532	64.6	5.5
MuonCandidate	3 559	21 443	16 110	58.0	8.6
ContainedPion	1 577	2 649	1 429	25.7	27.8
MuonIn3Planes	1 512	2 500	1 189	24.7	29.0
PionIn3Planes	1 409	2 258	905	23.0	30.8
ShowerCut	1 171	1 061	591	19.1	41.4
OpeningAngle	1 162	965	197	19.0	50.0
Nonprotons	1 081	746	136	17.6	55.0

RHC (Runs 1,2,3,4a-c; 1.108×10^{21} POT). MC prediction, per-run POT scaled; data blind.

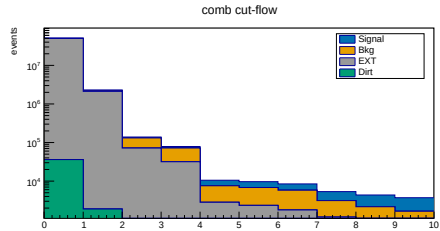


RHC stack (signal blue, bkg orange, EXT grey, dirt green; log scale).

Backup: inclusive cut-flow — combined (full exposure)

Cut	Signal	Bkg	EXT	Eff%	Pur%
None	11 710	583 527	49.8M	100.0	0.02
InFiducialVol	9 756	131 249	2.12M	83.3	0.43
Topological	7 461	59 669	71 717	63.7	5.4
MuonCandidate	6 702	40 423	31 627	57.2	8.5
ContainedPion	2 972	4 743	2 805	25.4	28.2
MuonIn3Planes	2 847	4 475	2 333	24.3	29.4
PionIn3Planes	2 648	4 028	1 777	22.6	31.3
ShowerCut	2 218	1 959	1 161	18.9	41.5
OpeningAngle	2 202	1 778	387	18.8	50.4
Nonprotons	2 055	1 389	266	17.5	55.4

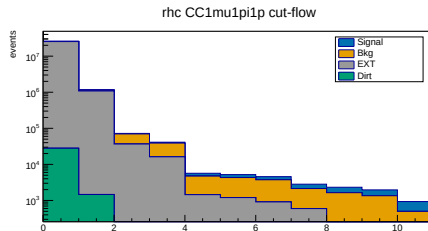
Combined (1.994×10^{21} POT). Efficiency/purity track the two beams closely — the selection is beam-independent.



Combined stack (log scale).

Backup: proton-tagged cut-flow — RHC (full exposure)

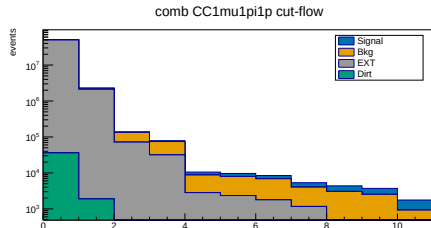
Cut	Signal	Bkg	EXT	Eff%	Pur%
None	3 638	306 840	25.3M	100.0	0.01
InFiducialVol	3 064	71 162	1.08M	84.2	0.26
Topological	2 455	33 134	36 532	67.5	3.4
MuonCandidate	2 234	22 769	16 110	61.4	5.4
ContainedPion	935	3 291	1 429	25.7	16.5
MuonIn3Planes	892	3 119	1 189	24.5	17.1
PionIn3Planes	831	2 835	905	22.9	18.2
ShowerCut	687	1 544	591	18.9	24.3
OpeningAngle	683	1 444	197	18.8	29.4
Nonprotons	611	1 217	136	16.8	31.1
IdentProton	436	443	55	12.0	46.6



RHC proton-tagged stack; last stage IdentProton (leading-proton tag). Purity 31.1% $\rightarrow$ 46.6%.

Backup: proton-tagged cut-flow — combined (full exposure)

Cut	Signal	Bkg	EXT	Eff%	Pur%
None	7 025	588 212	49.8M	100.0	0.01
InFiducialVol	5 898	135 106	2.12M	84.0	0.26
Topological	4 693	62 436	71 717	66.8	3.4
MuonCandidate	4 265	42 860	31 627	60.7	5.4
ContainedPion	1 787	5 928	2 805	25.4	16.9
MuonIn3Planes	1 704	5 618	2 333	24.3	17.6
PionIn3Planes	1 585	5 091	1 777	22.6	18.7
ShowerCut	1 320	2 857	1 161	18.8	24.7
OpeningAngle	1 310	2 670	387	18.7	30.0
Nonprotons	1 179	2 266	266	16.8	31.8
IdentProton	837	816	109	11.9	47.5



Combined proton-tagged stack; IdentProton purity
31.8% $\rightarrow$ 47.5%.

Backup: per-run exposure

Mode	Run	Data POT	MC/data scale
FHC	1	3.28×10^{20}	0.141
	2	1.27×10^{20}	0.051
	4	2.08×10^{20}	0.073
	5	2.23×10^{20}	0.116
RHC	1–4	1.11×10^{21}	0.045–0.091

Each run scaled by its own $D_{\text{run}}/\text{MC}_{\text{run}}$ (real-data-ready).

Backup: systematic breakdown (combined)

Source	p_μ	p_π	$\cos \theta_\mu$	$\cos \theta_\pi$	$\theta_{\mu\pi}$
Total	30.1	33.7	41.0	34.8	29.3
Flux (PPFX)	22.1	24.2	29.1	25.3	21.6
Detector	13.5	19.6	23.5	20.6	15.6
Cross section	9.4	8.5	9.9	8.6	7.6
Data stat	10.0	6.1	11.0	5.4	7.6

Bin-averaged fractional uncertainty [%]. Flux dominates throughout.

Backup: W/TKI observable definitions

From the muon, charged-pion and leading-proton momenta (STV tool, option 1):

$$W_{\pi p} = \sqrt{(E_\pi + E_p)^2 - |\vec{p}_\pi + \vec{p}_p|^2} \quad (\text{direct, no } \nu \text{ direction})$$

$$W_{\text{had}} = \sqrt{m_p^2 + 2m_p(E_{\text{cal}} - E_\mu) - Q^2} \quad (\text{calorimetric})$$

$$\delta p_T = |\vec{p}_T^\mu + \vec{p}_T^{\text{had}}|, \quad \delta\alpha_T, \delta\phi_T \text{ (TKI angles)}, \quad p_n = \sqrt{\delta p_T^2 + p_L^2}$$

Proton momentum from the proton-hypothesis range kinetic energy; $E_b = 24.78$ MeV argon removal energy.

Backup: proton-tagged binning & closure

Observable	Bin edges (6 equal-population bins)
$W_{\pi p}$	1.08, 1.19, 1.23, 1.27, 1.34, 1.47, 2.00 GeV/ c^2
W_{had}	0, 0.78, 1.08, 1.21, 1.31, 1.44, 1.90 GeV/ c^2
δp_T	0, 0.43, 0.59, 0.72, 0.88, 1.17, 1.60 GeV/ c
$\delta\alpha_T$	0, 80, 120, 142, 158, 170, 180 deg
$\delta\phi_T$	0, 37, 63, 84, 107, 139, 180 deg
p_n	0, 0.52, 0.67, 0.81, 0.98, 1.25, 1.70 GeV/ c

+ five kinematic observables ($p_\mu, p_\pi, \cos\theta_\mu, \cos\theta_\pi, \theta_{\mu\pi}$) on inclusive binning.

FHC W/TKI closure σ_{int} : 0.62, 0.60, 0.61, 0.67, 0.63, 0.52; $\chi^2/\text{ndf} \ll 1$, all positive.

RHC not yet validated: all 11 observables unfold negative (-0.48 to -0.04) — background $>$ data in this $\sim 47\%$ -purity subsample. Earlier positive RHC values came from an incorrect MC POT normalisation and are retracted. Comb deferred behind RHC.

Backup: generator predictions

- ▶ Inclusive: per-observable $d\sigma/dx$ from GENIE (gst), GiBUU (FinalEvents), NEUT (neutvect), NuWro (event output), flux-combined to per-argon 10^{-38} cm^2 .
- ▶ Proton-tagged W/TKI: same event samples reprocessed with the proton tag + TKI construction; NuWro and NEUT run inside the SL7 aptainer container.
- ▶ FTE rebinned to the coarsened analysis binning so every generator overlays on every observable.